Supplementary Material for Auxiliary Supervision in Graph Traffic Forecasting: A Controlled Study of Utility and Loss Effects

1 Scope, data construction and metric definitions

This supplement reports run-level results and diagnostic summaries. Benchmark comparisons, controlled JiNan experiments and controlled METR-LA experiments use distinct data protocols. The independent training unit is a random seed. Five diagnostic batches are nested within each of three seeds. The weight-sensitivity comparisons use seed 42.

JiNan controls use target-owned intervals $[0, 6249)$, $[6249, 7142)$ and $[7142, 8928)$, with training-prefix scaler mean 48.8709058474 and standard deviation 53.8879070877. Their node-specific weekday–slot reference is fitted on the training prefix with the recorded slot smoothing. METR-LA controls use $[0, 23990)$, $[23990, 27417)$ and $[27417, 34272)$; the scaler mean and standard deviation are 54.4059119157 and 19.4942512012. METR-LA references use node-specific positive-speed weekday–slot training means, with a node training-mean fallback for an unobserved slot. Target-crossing windows are excluded. This produces 23,967/3,416/6,844 METR-LA training/validation/test windows.

All-entry normalized MAE/RMSE and original-scale metrics on targets $y > 10^{-3}$ are distinct definitions. MAPE is $100\,\mathrm{mean}(|\hat{y} - y|/y)$ over valid original-scale entries. Public original-scale units are mph; JiNan original-scale values are flow. Different masks can change rankings even though scaling is affine. Validation selection uses the implemented normalized nonzero-label masked prediction loss, averaged over batches; test metrics never choose weights or checkpoints.

Public-dataset benchmark reference channels use observations through 3 June 2012 (METR-LA) or 24 May 2017 (PEMS-BAY). Their sliding windows are partitioned 70/10/20, so the historical references include validation-period observations. At three sampled test windows per dataset, reference values match those computed using these cutoff dates. Array checksums and sampled-window hashes identify the benchmark data files. Controlled groups construct reference channels using training observations only.

2 Run-level controlled JiNan results

The inventory includes 30 completed core runs under two warm-up settings. The S+P seed-456 factorial cell uses RTX 4090; all other warm-up-zero core runs use RTX 3090. Warm-up-20 core runs use RTX 4090. Two additional seed-42 runs at $\delta = 0.1$ reproduce the reported values to the displayed precision. Checkpoint hashes and metric-source records accompany the machine-readable inventory.

Table S1: Controlled JiNan inventory. MAE/RMSE normalized; MAPE original-scale percent. GPU column gives hardware family.

Configuration	Seed	e_w	GPU	MAE	RMSE	MAPE
Prediction-only	42	0	3090	0.154760	0.270620	33.821
P	42	0	3090	0.157192	0.274520	34.605
S+P	42	0	3090	0.157272	0.273686	34.534
TP+P	42	0	3090	0.157036	0.274388	34.674
Full	42	0	3090	0.156742	0.272582	34.421
P ($\delta = 0$)	42	0	3090	0.154494	0.270487	33.848
Full ($\delta = 0$)	42	0	3090	0.155743	0.270443	33.704
Prediction-only	123	0	3090	0.154868	0.270561	33.630
P	123	0	3090	0.156435	0.273361	34.401
S+P	123	0	3090	0.157223	0.274585	34.038
TP+P	123	0	3090	0.155905	0.272430	33.731
Full	123	0	3090	0.156921	0.273842	33.840

Configuration	Seed	e_w	GPU	MAE	RMSE	MAPE
P ($\delta = 0$)	123	0	3090	0.154743	0.270467	34.132
Full ($\delta = 0$)	123	0	3090	0.156140	0.272393	33.676
Prediction-only	456	0	3090	0.154323	0.269607	33.893
P	456	0	3090	0.157086	0.274614	34.284
S+P	456	0	4090	0.157791	0.274325	34.333
TP+P	456	0	3090	0.156066	0.273774	34.333
Full	456	0	3090	0.156632	0.272981	34.326
P ($\delta = 0$)	456	0	3090	0.153989	0.268978	33.776
Full ($\delta = 0$)	456	0	3090	0.155885	0.271816	34.076
P	42	20	4090	0.155562	0.271476	34.078
Full	42	20	4090	0.157376	0.274520	34.390
Full ($\delta = 0$)	42	20	4090	0.155827	0.272645	34.135
P	123	20	4090	0.157059	0.274986	34.534
Full	123	20	4090	0.157472	0.274592	34.236
Full ($\delta = 0$)	123	20	4090	0.156129	0.272139	34.633
P	456	20	4090	0.156812	0.274182	34.255
Full	456	20	4090	0.156658	0.273477	34.038
Full ($\delta = 0$)	456	20	4090	0.154867	0.271250	33.820

3 Paired effects and uncertainty

Paired differences are computed within seed before means and sample SDs. Metric-wise 95% t intervals use two degrees of freedom for three training seeds. Same-GPU subsets with two seeds are summarized descriptively. The factorial interaction residual is $I = E_F - E_{SP} - E_{TP} + E_P$. Deviation penalties are $D_P = E(P, on) - E(P, off)$ and $D_F = E(F, on) - E(F, off)$, with difference-in-differences $J = D_F - D_P$. Full-minus-P measures the overall joint forecasting gain.

Table S2: Core paired effects in normalized MAE, with metric-wise 95% t intervals. Mixed hardware estimates are exploratory; two-seed rows omit intervals.

Comparison	e_w	n	Mean	SD	CI lower	CI upper	GPU
P minus Prediction-only	0	3	0.002254	0.000617	0.000721	0.003788	3090
S+P minus Prediction-only	0	3	0.002779	0.000603	0.001281	0.004276	mixed
S+P minus Prediction-only	0	2	0.002434	0.000111	–	–	3090 (2)
TP+P minus Prediction-only	0	3	0.001686	0.000621	0.000142	0.003229	3090
Full minus Prediction-only	0	3	0.002114	0.000172	0.001686	0.002543	3090
S+P minus P	0	3	0.000525	0.000387	-0.000436	0.001485	mixed
S+P minus P	0	2	0.000434	0.000500	–	–	3090 (2)
TP+P minus P	0	3	-0.000569	0.000434	-0.001646	0.000509	3090
Full minus P	0	3	-0.000140	0.000541	-0.001484	0.001205	3090
Full minus TP+P	0	3	0.000429	0.000666	-0.001225	0.002083	3090
Full minus S+P	0	3	-0.000664	0.000444	-0.001767	0.000438	mixed
Full minus S+P	0	2	-0.000417	0.000162	–	–	3090 (2)
Interaction I	0	3	-0.000096	0.000304	-0.000850	0.000659	mixed
Interaction I	0	2	-0.000074	0.000426	–	–	3090 (2)
Full minus P	20	3	0.000691	0.001013	-0.001825	0.003207	4090
Full ($\delta = 0$) minus Full	20	3	-0.001561	0.000225	-0.002119	-0.001003	4090
D_P	0	3	0.002496	0.000724	0.000699	0.004293	3090
D_F	0	3	0.000842	0.000137	0.000502	0.001182	3090
Interaction J	0	3	-0.001654	0.000720	-0.003443	0.000136	3090

4 Nonzero deviation-weight sensitivity

The candidate set is $\{0.03, 0.1, 0.3\}$ for P and Full at seed 42. All candidates share the validation criterion, 200-epoch maximum budget and patience 30. Delta 0.1 uses existing models; four additional runs evaluate the other candidates. Minimum validation prediction loss selects 0.03 for each topology. The table reports every candidate.

Table S3: JiNan weight sensitivity at seed 42. MAE/RMSE normalized; MAPE original-scale percent. Asterisk denotes validation-selected weight.

Model	δ	Val MAE	Test MAE	Test RMSE	MAPE
Full*	0.03	0.148694	0.155571	0.270356	34.244
P*	0.03	0.149170	0.155202	0.271300	33.801
Full	0.1	0.150042	0.156742	0.272582	34.421
P	0.1	0.151883	0.157192	0.274520	34.605
Full	0.3	0.151628	0.157305	0.273565	34.222
P	0.3	0.151539	0.156794	0.274382	34.516

5 Matched METR-LA results

The twelve controls share a training-prefix-only protocol, RTX 3090, seed-matched settings, a 200-epoch maximum budget and patience 30. Six original runs use $\delta = 1$; six paired interventions set $\delta = 0$ and retain every other scientific setting. For Full/seed456 at $\delta = 1$, early stopping occurs at epoch 158 and validation selects epoch 128. At $\delta = 0$, Prototype/seed42 and Joint/seed123 complete 200 epochs. The other four interventions stop after 30 epochs without validation improvement, at epochs 131, 92, 86 and 179. Physical metrics and all-entry normalized metrics are reported separately.

Table S4: METR-LA test metrics from independently replayed validation-selected models. Physical MAE/RMSE in mph; MAPE in percent.

Seed	Model	δ	MAE	RMSE	MAPE	Norm MAE	Norm RMSE
42	Prototype	1	4.1054	9.8354	10.7304	0.241014	0.624565
42	Prototype	0	4.0152	9.7330	10.3167	0.233649	0.614204
42	Joint	1	4.1119	9.8064	10.6604	0.240926	0.621577
42	Joint	0	4.0657	9.7984	10.5082	0.236754	0.618522
123	Prototype	1	4.2407	10.1580	10.9975	0.244136	0.626997
123	Prototype	0	4.1116	9.9318	10.6970	0.236897	0.617219
123	Joint	1	4.1004	9.7538	10.6552	0.241432	0.622150
123	Joint	0	4.0784	9.8360	10.5525	0.237777	0.620716
456	Prototype	1	4.2546	10.1843	11.0218	0.243898	0.627474
456	Prototype	0	4.1401	9.9750	10.7060	0.238024	0.619045
456	Joint	1	4.1761	9.8311	10.8840	0.243515	0.618993
456	Joint	0	4.0589	9.6968	10.5641	0.237880	0.614173

Table S5: METR-LA paired summaries ($n = 3$). Joint-minus-Prototype uses $\delta = 1$. Penalties are On minus Off; $J = D_F - D_P$. Intervals are metric-wise descriptive 95% t intervals (df2), without multiplicity-adjusted significance claims.

Comparison	Metric	Mean	SD	CI lower	CI upper
Joint minus Prototype	Normalized MAE	-0.001059	0.001433	-0.004618	0.002500
Joint minus Prototype	Normalized RMSE	-0.005439	0.002794	-0.012379	0.001502
Joint minus Prototype	Physical MAE (mph)	-0.070757	0.073712	-0.253868	0.112355
Joint minus Prototype	Physical RMSE (mph)	-0.262147	0.203513	-0.767701	0.243408
Joint minus Prototype	MAPE (percentage points)	-0.183396	0.141764	-0.535557	0.168765
D_P	MAE (mph)	0.111259	0.019614	0.062534	0.159983
D_P	RMSE (mph)	0.179293	0.067179	0.012412	0.346175
D_P	MAPE (pp)	0.343354	0.061424	0.190768	0.495939
D_P	Normalized MAE	0.006826	0.000827	0.004772	0.008881
D_P	Normalized RMSE	0.009523	0.000991	0.007061	0.011985
D_F	MAE (mph)	0.061795	0.049503	-0.061177	0.184768
D_F	RMSE (mph)	0.019997	0.108712	-0.250059	0.290052
D_F	MAPE (pp)	0.191535	0.113860	-0.091310	0.474380
D_F	Normalized MAE	0.004487	0.001027	0.001936	0.007039
D_F	Normalized RMSE	0.003103	0.001694	-0.001104	0.007310

Comparison	Metric	Mean	SD	CI lower	CI upper
J	MAE (mph)	-0.049463	0.055179	-0.186537	0.087610
J	RMSE (mph)	-0.159297	0.129466	-0.480909	0.162315
J	MAPE (pp)	-0.151819	0.138711	-0.496396	0.192759
J	Normalized MAE	-0.002339	0.001829	-0.006883	0.002205
J	Normalized RMSE	-0.006420	0.002489	-0.012603	-0.000237

6 Gradient and representation probe variability

We probe three warm-up-zero Full checkpoints selected at epochs 190, 195 and 200 for seeds 42, 123 and 456. Each seed uses five fixed, disjoint training batches of 16 windows. The same windows and augmentation seeds are reused across seeds. Probes use training mode with curriculum sampling disabled and model parameters fixed. Parameter and buffer states are checked after each probe. Gradients cover all encoder parameters before clipping and Adam. Combined ratios use the gradient of the weighted summed auxiliary loss; the ramp is one at these checkpoints. The gradient equation in main-text Section 4.5 defines the encoder-gradient ratio and cosine. The ratio denominator is floored at 10^{-12} in code; cosines are undefined when either norm is at most 10^{-12}. Near-zero gradient norms are reported as undefined cosines. SD denotes batch variability within each training seed.

Table S6: Checkpoint-probe gradient summaries over five batches within each seed. SD denotes within-seed batch variability.

Seed	Loss	Cos mean	Cos SD	Ratio mean	Ratio SD
42	Combined	0.0444	0.1029	3.3892	1.2038
42	Spatial	-0.0031	0.1087	1.0404	0.4608
42	Temporal	0.0225	0.1258	0.0280	0.0082
42	Contrastive	-0.1053	0.2373	0.0634	0.0211
42	Deviation	0.0466	0.0871	3.1757	1.2248
123	Combined	-0.0106	0.0701	3.7244	1.8120
123	Spatial	0.0143	0.0820	3.2574	1.5825
123	Temporal	0.0488	0.1102	0.0464	0.0103
123	Contrastive	-0.0009	0.0731	0.0590	0.0132
123	Deviation	-0.0328	0.0248	2.1046	0.7836
456	Combined	-0.0300	0.1126	5.9867	2.6478
456	Spatial	0.0475	0.1957	2.5580	1.5630
456	Temporal	-0.1122	0.0906	0.1327	0.0640
456	Contrastive	-0.0859	0.1227	0.0562	0.0285
456	Deviation	-0.0451	0.0701	5.3945	2.3081

Effective rank is the exponential entropy of normalized singular values of the centered window–node by feature matrix. Constant matrices have effective rank zero. Node variance is computed across nodes within each window; sample variance is across windows for each node. Relative view distance divides each window–node difference norm by the original-state norm floored at 10^{-8}. Pearson correlations summarize dependent window–node observations within each batch. All probes use the Full configuration with temporal prediction.

Table S7: Step-12 representation summaries over five batches per seed. Feature dimension is 64; rank SD denotes within-seed batch variability.

Seed	View	Rank mean	Rank SD	Node variance	Sample variance
42	Original	31.179	0.436	0.32079	0.24181
42	Historical	28.978	0.518	0.26733	0.18024
42	Spatial	33.637	0.406	0.31488	0.22987
123	Original	31.567	0.206	0.30034	0.17178
123	Historical	29.568	0.229	0.24988	0.11799
123	Spatial	33.530	0.160	0.29950	0.16848
456	Original	31.785	0.248	0.29742	0.22220
456	Historical	29.716	0.363	0.24366	0.16437
456	Spatial	33.818	0.295	0.29361	0.21554

Table S8: Step-12 relative view distance and descriptive association with forecasting error. Pearson limits denote the observed range across five batches.

Seed	View	Mean distance	Pearson min	Pearson max
42	Spatial	0.1306	-0.0454	0.0143
42	Historical	0.4615	0.0315	0.1804
123	Spatial	0.0858	-0.0595	0.0405
123	Historical	0.4034	0.1705	0.2355
456	Spatial	0.0976	-0.0461	0.0381
456	Historical	0.4408	0.1488	0.2927

7 Objective implementation and historical settings

The published ST-SSDL triplet uses squared distances with a detached query; its released code uses unsquared distances with a detached query. Its deviation objective matches detached-query distances to prototype distances. UPSTCN instead uses an unsquared triplet with query gradients and aligns historical queries and prototypes against stop-gradient current targets. The intervention toggles this historical-alignment term. These distinctions complement the cited objective definitions in the main text. The checkpoint-matched historical ST-SSDL baseline scripts retain the detached-query unsquared triplet and distance-of-distances deviation objective. The benchmark adaptation changes data preprocessing and loading, while the historical-alignment objective belongs to UPSTCN.

The JiNan benchmark prototype-based models use a network-aggregated historical reference broadcast to sensors. HA and the controlled groups use sensor-specific historical references computed from training observations. These protocol groups are analyzed separately.

The warm-up-zero controls, weight-sensitivity runs and matched METR-LA comparisons use RTX 3090, except for the S+P seed-456 factorial run on RTX 4090. Warm-up-20 controls and trajectory diagnostics use RTX 4090. Multi-seed checkpoint probes use RTX 3090. Configuration files in the source-data index specify learning-rate schedules and all objective weights. METR-LA controls use warm-up zero, deviation weight one and learning-rate multipliers of 0.1 at epochs 50 and 70.

Table S9: Benchmark training settings (panel a). LR is initial learning rate; decay lists milestones. Warm-up is specific to the UPSTCN activation schedule; a dash in that column means not applicable. A decay dash means no milestone schedule is recorded. NR denotes an unrecorded fixed seed. AGCRN settings are taken from the reference configuration files.

Data/model	Seed	Batch	LR	Warm-up	Decay
JiNan / GWNet	NR	64	0.001	–	–
JiNan / AGCRN	42	16	0.003	–	50/100
JiNan / TP	42	16	0.001	20	50/100
JiNan / S+TP	42	16	0.001	20	50/100
JiNan / ST-SSDL	42	16	0.001	–	50/100
METR-LA / GWNet	NR	64	0.001	–	–
METR-LA / AGCRN	42	64	0.003	–	50/100
METR-LA / TP	42	64	0.001	0	50/70
METR-LA / S+TP	42	64	0.001	0	50/70
METR-LA / ST-SSDL	42	64	0.001	–	50/70
PEMS-BAY / GWNet	NR	64	0.001	–	–
PEMS-BAY / AGCRN	42	64	0.003	–	50/100
PEMS-BAY / TP	42	64	0.001	0	10/70/150
PEMS-BAY / ST-SSDL	666	64	0.001	–	10/70/150

JiNan benchmark UPSTCN and AGCRN entries and all Graph WaveNet entries use 150-epoch budgets; ST-SSDL and the other public-dataset entries use 200.
Panel (b): joint-model benchmark results. JiNan MAE/RMSE are normalized; METR-LA MAE/RMSE are in mph. MAPE is in percent.

Dataset	MAE	RMSE	MAPE
JiNan	0.1610	0.2793	35.03
METR-LA	3.1030	6.3941	8.77

8 Offline DIM diagnostic

We evaluate one DIM checkpoint selected by validation loss (training seed 42, epoch 149). Its validation and test metrics match the reported values to their stated precision. Five fixed training batches contain 16 windows each.

Forward probes retain the original batch-permutation negative sampler, including permutation fixed points, and the sigmoid-of-node-mean readout. Model parameters and buffers remain unchanged throughout these probes.

The discriminator has BCE near $\ln 2$, AUROC near 0.5 and balanced accuracy near 0.5 across the five batches. Positive and negative predicted probabilities both concentrate near 0.5. Nonzero node variances and effective ranks characterize the corresponding encoder states. The five batches describe within-checkpoint variation from a single training seed. Training of this DIM checkpoint excluded batches with nonfinite losses. Raw logits, checkpoint identity and source hashes accompany the diagnostic records.

Table S10: DIM checkpoint diagnostic over five fixed batches. Variance is across nodes; rank is the effective rank. Original and augmented views use the same checkpoint. Balanced accuracy uses a probability threshold of 0.5.

Batch	BCE	AUROC	Bal. acc.	Orig. var.	Aug. var.	Orig. rank	Aug. rank
1	0.693147	0.5000	0.4986	0.5123	0.5026	26.63	30.25
2	0.693147	0.5001	0.5005	0.5430	0.5322	28.40	31.01
3	0.693147	0.4998	0.5009	0.5152	0.5074	27.83	30.75
4	0.693147	0.5004	0.5007	0.4947	0.4897	27.38	30.48
5	0.693147	0.5001	0.4984	0.5470	0.5319	28.27	31.55

9 Source data and reproducibility

The source-data tables contain metrics, paired differences, sample SDs, selection records and checkpoint identities. Core results appear in `verified_result_inventory.csv`. Independent replays and sensitivity results appear in `all_recomputed_metrics.csv` and `jinan_weight_sensitivity.csv`. Probe records identify seed, batch, recurrent step and view. The source-data index links analysis scripts and manifests. Access follows the manuscript's data-availability statement.

Source hashes identify the trainer version used for each run. An epoch-checkpoint update adds model, optimizer, scheduler and random-state persistence while retaining the same loss definitions and forward computation. The first P/delta-0.03 sensitivity run predates this update; later sensitivity runs use the revised checkpoint handling. Source hashes and consistency tests accompany the code records.